

Gate-Induced Interfacial Superconductivity in 1T-SnSe₂

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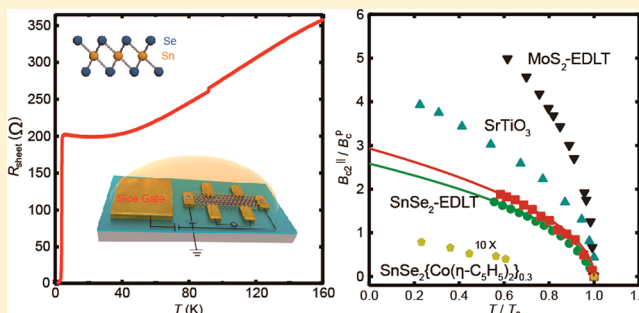
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Supporting Information

ABSTRACT: Layered metal chalcogenide materials provide a versatile platform to investigate emergent phenomena and two-dimensional (2D) superconductivity at/near the atomically thin limit. In particular, gate-induced interfacial superconductivity realized by the use of an electric-double-layer transistor (EDLT) has greatly extended the capability to electrically induce superconductivity in oxides, nitrides, and transition metal chalcogenides and enable one to explore new physics, such as the Ising pairing mechanism. Exploiting gate-induced superconductivity in various materials can provide us with additional platforms to understand emergent interfacial superconductivity. Here, we report the discovery of gate-induced 2D superconductivity in layered 1T-SnSe₂, a typical member of the main-group metal dichalcogenide (MDC) family, using an EDLT gating geometry. A superconducting transition temperature $T_c \approx 3.9$ K was demonstrated at the EDL interface. The 2D nature of the superconductivity therein was further confirmed based on (1) a 2D Tinkham description of the angle-dependent upper critical field B_{c2} , (2) the existence of a quantum creep state as well as a large ratio of the coherence length to the thickness of superconductivity. Interestingly, the in-plane B_{c2} approaching zero temperature was found to be 2–3 times higher than the Pauli limit, which might be related to an electric field-modulated spin–orbit interaction. Such results provide a new perspective to expand the material matrix available for gate-induced 2D superconductivity and the fundamental understanding of interfacial superconductivity.

KEYWORDS: SnSe₂, metal dichalcogenides, electric-double-layer transistor, interfacial superconductivity



Exploration of two-dimensional (2D) superconductivity in new material systems has attracted intensive research interest in condensed matter physics.^{1–4} In particular, EDL gating has been utilized as a powerful technique to realize interfacial carrier accumulation at an ultrahigh density level for realization of 2D superconductivity in various semiconductors and insulators, such as transition metal dichalcogenides (TMDCs),^{5–13} ZrNCl,^{14,15} and perovskite oxides (e.g., SrTiO₃ and KTaO₃).^{16,17} In these systems, emergent interfacial phenomena have been observed, such as the 2D Ising electron pairing protected by spin-valley locking in MoS₂, which induces the enhancement of the Pauli paramagnetic limit,^{8,9} and the suppression of charge density waves in favor of superconductivity in TiSe₂, providing an electrical method to control the many-body interaction.⁷

Compared with TMDCs that show rich electronic properties typically due to the versatile electron filling of heavy d-orbitals of the transition metals,^{18–20} the electronic structures of main-group MDCs can be quite different.^{21–24} For example, a typical main-group layered semiconductor, InSe, with conduction and valence band edges mainly contributed by s- and p-orbital electrons, was reported to host a small effective mass and high electron mobility, leading to demonstration of the quantum Hall state.^{25,26} Previous studies of electric field-tuned layered main-group MDCs have focused on normal-state properties, such as the insulator–metal transition in SnS₂²⁷ and the quantum Hall effect in InSe.²⁵ Gating-induced 2D super-

Received: December 7, 2017

Revised: January 23, 2018

Published: January 31, 2018

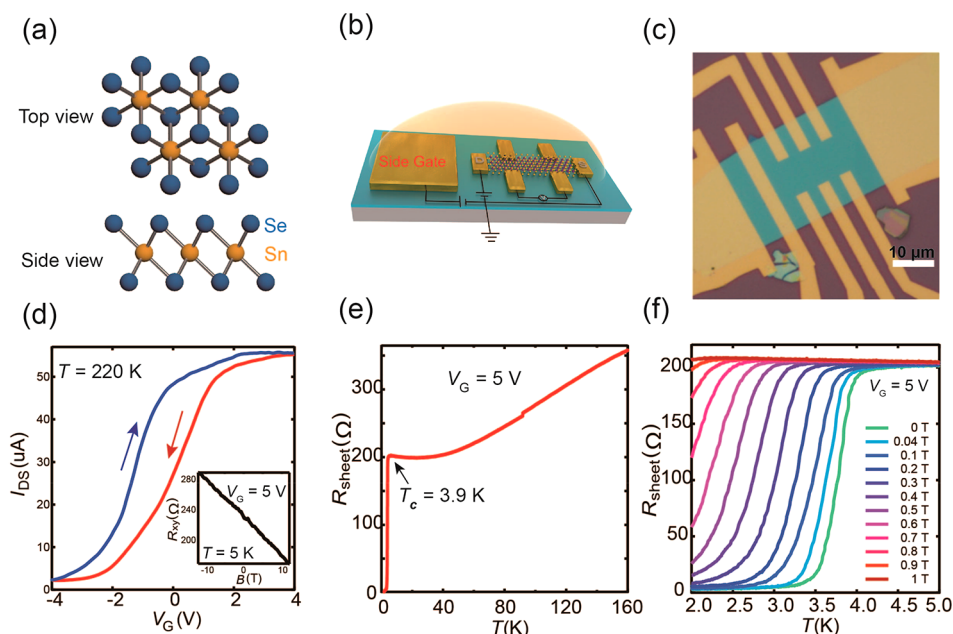


Figure 1. Crystal structure of 1T-SnSe₂, transfer curve, and superconductivity of SnSe₂-EDLT. (a) The top and side views of the crystal structure. (b) Sketch of a SnSe₂-EDLT device. (c) Optical microscope image of a typical multiterminal Hall bar device. The scale bar is 10 μm . (d–f) Transfer curve and superconductivity of dev #01. (d) I_{ds} as a function of sweeping V_{G} with $V_{\text{sd}} = 0.1$ V. The inset shows the magnetic field dependence of R_{xy} at $T = 5$ K with $V_{\text{G}} = 5$ V. (e) Temperature-dependent resistance between 2 and 160 K with $V_{\text{G}} = 5$ V. A metal-to-superconductor transition occurs at $T_{\text{c}} = 3.9$ K. (f) Temperature dependence of the magnetoresistance. The measurements were carried out using an excitation current of 50 nA and varying out-of-plane magnetic fields up to 1 T under the same V_{G} .

conductivity in the main-group MDCs has not yet been observed. SnSe₂, one of main-group MDCs materials, has been reported to support superconducting state by organometallic intercalation.^{28–32} The question about whether its intrinsic 2D superconductivity in the few-layer SnSe₂ exists is worth to be explored.

In this study, we demonstrated a gate-induced superconductivity with $T_{\text{c}} \approx 3.9$ K in mechanically exfoliated 1T-SnSe₂ based on the ionic liquid gating technique. Angle and temperature-dependent $B_{\text{c}2}$ measurements are used to show the 2D nature of the superconducting state. The temperature dependence of the resistance in out-of-plane magnetic fields reveals the nature and dynamics of the vortex state below T_{c} . The in-plane $B_{\text{c}2}$ measurements suggest a spin–orbit interaction (SOI)-related depairing mechanism.

The SnSe₂ studied here is a layered semiconductor with a 1T-type crystal structure.²⁴ As shown in Figure 1a (with top and side views), the Sn layer is sandwiched between two Se layers in monolayer 1T-SnSe₂ with ABC (Se–Sn–Se) atomic stacking. A Sn atom is surrounded by six Se atoms, forming an octahedral coordination.³³ Micro-Raman spectroscopy and transmission electron microscopy (TEM) were employed to understand the crystalline properties of the mechanically exfoliated thin flakes. The Raman spectra measured using a 514 nm laser at room temperature showed two sharp peaks, as shown in Figure S1c (the out-of-plane $A_{1\text{g}}$ mode located at 184 cm^{-1} and in-plane E_{g} mode located at 110 cm^{-1}). Similar to other reports, such an observation indicates a 1T-type crystal phase for the SnSe₂ single crystals.^{34,35} The high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) image is shown in Figure S1d; the observed arrangement of Sn and Se atoms agrees well with the top view shown in Figure 1a.

Figure 1b shows a schematic illustration of a SnSe₂-EDLT device with an ionic liquid gate (DEME-TFSI). The side gate drives corresponding ions toward the channel surface, resulting in the formation of an EDL capacitance with an ultrathin (~ 1 nm) Helmholtz layer.^{15,17,19} As a result, a high carrier density can be accumulated at the surface of the thin flakes. Figure 1c shows the optical image of a typical SnSe₂ device (see fabrication details in Methods). We first measured the transfer characteristics of the SnSe₂-EDLT at 220 K (above the glass transition temperature of the ionic liquid) with a current ON/OFF ratio of ~ 25 , as shown in Figure 1d.^{5,9,36} The application of a side gate voltage of $V_{\text{G}} = 5$ V leads to the buildup of a sheet carrier density as high as $1.4 \times 10^{14} \text{ cm}^{-2}$ at $T = 5$ K (inset of Figure 1d). In this paper, the data of superconductivity are from four typical devices. Note that the ionic liquid gating technique may induce inhomogeneous carrier doping across the surface of SnSe₂ samples, leading to subtle different superconductivity performance in different geometric devices, as pointed out in the prior work.⁶

The temperature-dependent sheet resistance curve ($R_{\text{sheet}} - T$) of a typical device with $V_{\text{G}} = 5$ V is shown in Figure 1e. The data show a metallic state during cooling to 4 K, with a sharp drop in resistance observed at temperatures below 4 K that is associated with the emergence of superconductivity. We estimated the critical temperature T_{c} to be approximately 3.9 K (T_{c} is defined as the temperature at which R_{sheet} is equal to $90\% \times R_{\text{n}}$, where R_{n} is the normal-state resistance). Figure 1f shows the $R_{\text{sheet}} - T$ curves for varying out-of-plane magnetic fields. T_{c} decreases with increasing magnetic field until the superconducting state is completely suppressed at approximately 1 T. Out-of-plane magnetoresistance measurements at various temperatures and the $V - I$ curves measured under different out-of-plane magnetic fields are shown in the Supporting Information. At higher temperatures, the super-

conductivity can be suppressed by a smaller magnetic field. The color plot of V versus I and B shows that the critical current is diminished by the increasing magnetic field (details shown in Figure S3).

Angle-dependent magnetoresistance measurements were also carried out to confirm the 2D nature of the observed superconducting state. Figure 2a shows the magnetoresistance

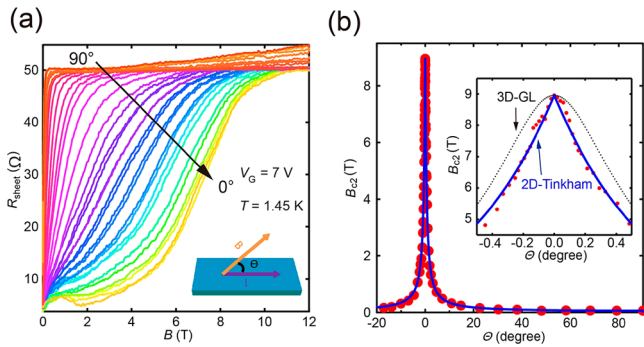


Figure 2. Confirmation of 2D superconductivity. (a) The magnetoresistance of dev #02 measured at different angles at $T = 1.45$ K and $V_G = 7$ V. The inset shows a schematic description of the angle θ between magnetic field and flake surface. (b) The B_{c2} as a function of θ extracted from (a). The inset shows B_{c2} (symbols) plotted in the range $\theta = \pm 0.5^\circ$. The blue solid line and the black dashed line are fits to the 2D Tinkham formula and 3D Ginzburg–Landau anisotropic model, respectively. The B_{c2} curve agrees well with the 2D Tinkham model.

(with $T = 1.45$ K and applied gate voltage $V_G = 7$ V) at various θ (defined as the angle between the magnetic field and flake surface, as illustrated in the inset of Figure 2a); the magnetoresistance is highly θ -dependent. Here, the magnetoresistance at approximately $\theta = 0^\circ$ was found to decrease with increasing magnetic field between 1 and 2 T. This observation is similar to a previous study of the interface of $\text{LaAlO}_3/\text{SrTiO}_3$ and amorphous Pb films that showed enhancement of superconductivity induced by an in-plane magnetic field over a small range, which may be related to the strong SOI.³⁷ We defined B_{c2} to be the critical magnetic field at which the resistance reaches $90\% \times R_n$, and plotted the measured B_{c2} versus θ in Figure 2b. B_{c2} is approximately 0.06 T for $\theta = 90^\circ$ and reaches a value as high as 9.0 T for $\theta = 0^\circ$. We fitted the B_{c2} - θ curves using both the 2D Tinkham formula, $\left(\frac{B_{c2}(\theta)\cos(\theta)}{B_{c2}^{\parallel}}\right)^2 + \left|\frac{B_{c2}(\theta)\sin(\theta)}{B_{c2}^{\perp}}\right| = 1$, and three-dimensional (3D) Ginzburg–Landau anisotropic model, $\left(\frac{B_{c2}(\theta)\cos(\theta)}{B_{c2}^{\parallel}}\right)^2 + \left(\frac{B_{c2}(\theta)\sin(\theta)}{B_{c2}^{\perp}}\right)^2 = 1$ (where $B_{c2}^{\parallel}$ and $B_{c2}^{\perp}$ are the upper critical fields at $\theta = 0^\circ$ and 90° , respectively).^{8,9,14} The inset of Figure 2b clearly shows that the fitting curve agrees well with the 2D Tinkham model, indicating the 2D nature of the observed superconductivity.

To understand the microscopic vortex state in the 2D superconductor,^{14,38,39} we plot the $R_{\text{sheet}}-T$ curves at different out-of-plane magnetic fields B (Figure 1f) using the Arrhenius convention in the Figure 3a. We note that for temperatures immediately below the onset of the resistance drop, the observed $R_{\text{sheet}}-T$ behavior is consistent with a thermally activated process: the temperature is comparable to the finite energy barriers expected for the motion of vortices with the vortices participating in a thermally activated flux flow

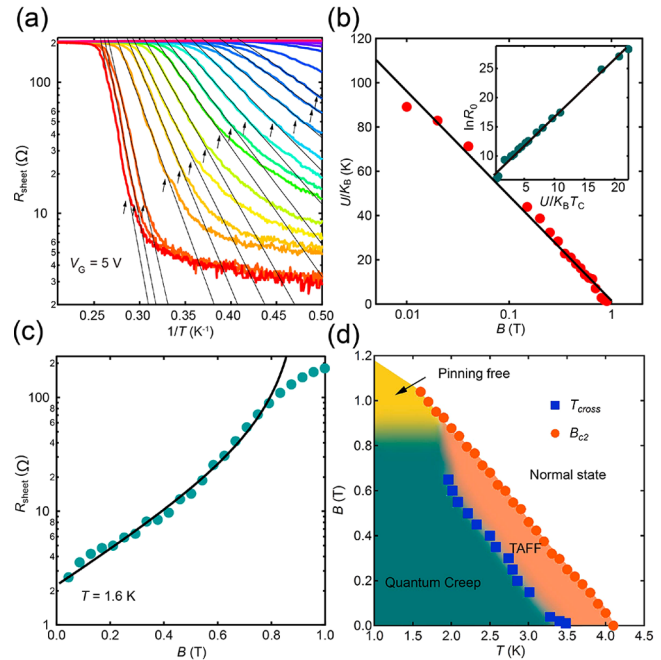


Figure 3. Vortex dynamics analysis. (a) An Arrhenius plot for the temperature dependence of the sheet resistance measured at different magnetic fields based on dev #01 at $V_G = 5$ V, which indicates thermally activated behavior (black solid lines). (b) Magnetic field dependence of the activation energy ($U(B)/k_B$), plotted on a semilogarithmic scale. The activation energy is well described by the equation $U(B) = U_0 \ln(B_0/B)$ (the black solid line). The inset shows the curve of $\ln R_0$ versus $U(B)/k_B T_c$. (c) The measured sheet resistance as a function of magnetic field at 1.6 K, which is fitted well by a quantum creep model below $B = 0.8$ T (the solid black line). (d) Vortex phase diagram of the SnSe_2 -EDLT device. The solid circles show B_{c2} at different temperatures. The solid squares T_{cross} are derived from the crossing points of the $R_{\text{sheet}} - 1/T$ curves with the thermal activation fitting curves shown in (a) (black arrows).

(TAF).^{40,41} The black lines are fits to a model of thermally activated behavior described by $R_{\text{sheet}} = R_0(B)\exp(-U(B)/k_B T)$, where $U(B)$ is the activation energy and k_B is the Boltzmann constant. The magnetic-field dependence of the extracted $U(B)/k_B$ is plotted in Figure 3b, whereas $R_0(B)$ as a function of $U(B)/k_B T_c$ is plotted in the inset. These two curves can be well fitted using the equations $U(B) = U_0 \ln(B_0/B)$ and $\ln(R_0(B)) = U(B)/k_B T_c + A$ (A is a constant), respectively, where U_0 is the vortex–antivortex binding energy. From the $U(B)$ fitting, we obtained $U_0/k_B = 20.6$ K and $B_0 = 1.1$ T, which are comparable to values obtained in the previous studies of ZrNCl and NbSe_2 .^{14,39}

Interestingly, the applied magnetic field gives rise to an apparent metallic ground state, as indicated by the saturating finite resistance as one approaches zero temperature. Note that this is different from the conventional magnetic field-induced superconductor–insulator transition in disordered systems.^{42,43} We show that the magnetic field dependence of the metallic ground state resistance follows a quantum creep model, which describes the temperature-independent quantum tunneling of vortices in the following equations^{14,44}

$$R_{\text{sheet}} \sim \frac{\hbar}{4e^2} \frac{\mathcal{L}}{1 - \mathcal{L}} \quad (1)$$

$$\mathcal{L} \sim \exp \left\{ C \frac{\hbar}{e^2} \frac{1}{R_N} \left(\frac{B - B_{c2}}{B_{c2}} \right) \right\} \quad (2)$$

where $\hbar$ is the reduced Planck constant, e is the elementary charge, and C is a dimensionless constant. As shown in Figure 3c, at $T = 1.6$ K, the magnetic field dependence of R_{sheet} can be described by a quantum creep model below 0.8 T, indicating that the vortices move via quantum creep. We further plot the magnetic field-temperature vortex phase diagram in Figure 3d. TAFF and quantum creep states are separated by T_{cross} (annotated by blue squares), which is defined as the point (indicated by the black arrows in Figure 3a) at which the thermal activation fitting curve deviates from the measured data. The system passes through a magnetic field driven superconductor-metal transition while approaching zero temperature.

To shed light on the unique characteristics of the electric field-induced 2D superconductivity in main-group MDCs, we also measured the temperature dependence of the magnetoresistance in both in-plane and out-of-plane magnetic fields. We can extract the coherence length and thickness of the superconductor by fitting the in-plane and out-of-plane B_{c2} with the 2D Ginzburg–Landau model

$$B_{c2}^{\perp}(T) = \frac{\Phi_0}{2\pi\xi_{\text{GL}}(0)^2} \left(1 - \frac{T}{T_c} \right) \quad (3)$$

$$B_{c2}^{\parallel}(T) = \frac{\Phi_0\sqrt{12}}{2\pi\xi_{\text{GL}}(0)d_{\text{sc}}} \left(1 - \frac{T}{T_c} \right)^{1/2} \quad (4)$$

where Φ_0 denotes the magnetic flux quantum (2.07×10^{-15} Wb), $\xi_{\text{GL}}(0)$ is the in-plane 2D GL coherence length at $T = 0$ K and d_{sc} is the thickness of the superconductor. On the basis of the analysis shown in Figure S5, we obtained $\xi_{\text{GL}}(0) \approx 53.7$ nm and $d_{\text{sc}} \approx 1.6$ nm. Note that the value for d_{sc} is close to the thickness of bilayer SnSe₂ (the interlayer distance of SnSe₂ is 0.62 nm³⁴) and $d_{\text{sc}} \ll \xi_{\text{GL}}(0)$, which clearly indicate the 2D nature of our system.

We now discuss the pairing mechanism in this system. In a conventional superconductor, a Cooper pair is composed of two electrons with opposite spin. In an external magnetic field, the superconducting state will be destroyed if the magnetic energy required for flipping the electron spin is comparable to the superconducting energy gap. The corresponding magnetic field is known as the Pauli paramagnetic limit B_c^{P} , which is given by $B_c^{\text{P}} \approx 1.85 \times T_c$ assuming the weak coupling limit of superconductivity.^{45,46} If we plot the normalized $B_{c2}^{\parallel}/B_c^{\text{P}}$ as a function of temperature (Figure 4) for the superconducting SnSe₂ devices, we can see that $B_{c2}^{\parallel}$ exceeds B_c^{P} by a factor of 2–3 for temperatures approaching 0 K. The enhancement of the Pauli paramagnetic limit under an in-plane magnetic field has been found in many 2D superconductors with a strong SOI effect.^{8,9,47,48} By application of a gate voltage and further breaking of the inversion symmetry, the SOI in SnSe₂ can be electrically modulated, as confirmed by magnetoresistance measurements (details shown in Supporting Information Figure S6). This also implies that the strong SOI induced by the perpendicular electric field plays an important role in the enhancement of the Pauli paramagnetic limit in SnSe₂. We also note that in bulk superconducting SnSe₂ with a lamellar organometallic intercalation, where the orbital effect dominates the pair-breaking mechanism of the superconductivity under a

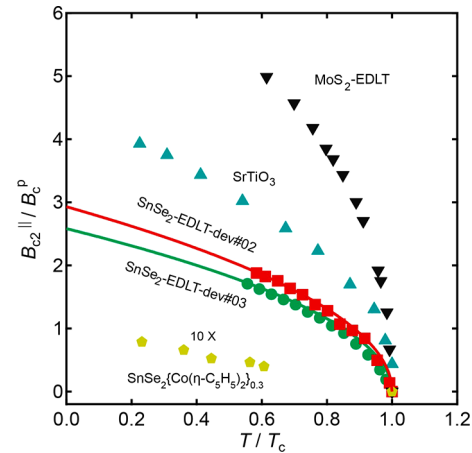


Figure 4. Temperature dependence of $B_{c2}^{\parallel}/B_c^{\text{P}}$ of two SnSe₂-EDLT devices (red square symbols (dev #02) and green circle symbols (dev #03)), MoS₂-EDLT (black triangles) (ref 8), SrTiO₃ (blue triangles) (ref 48), and SnSe₂{Co(η -C₅H₅)₂}_{0.3} (yellow pentagons) (ref 32). The values shown for SnSe₂{Co(η -C₅H₅)₂}_{0.3} were expanded 10-fold. The red and green lines denote the fitting curves obtained from the 2D Ginzburg–Landau models.

magnetic field, the $B_{c2}^{\parallel}$ is far smaller than the Pauli paramagnetic limit.³² However, in the 2D limit where $d_{\text{sc}} \ll \xi_{\text{GL}}(0)$ the orbital depairing effect is dramatically suppressed in the SnSe₂-EDLT. For comparison, we plotted the data reported for a MoS₂-EDLT⁸ and SrTiO₃⁴⁸ as well in Figure 4.

In summary, we have induced a superconducting state below $T_c = 3.9$ K in SnSe₂ by using the ionic liquid gating technique. The 2D nature of the superconducting state is confirmed by both the angle-dependent upper critical field B_{c2} described by 2D Tinkham and the fact that $d_{\text{sc}} \ll \xi_{\text{GL}}(0)$. The in-plane B_{c2} of the devices greatly exceeds the Pauli limit, which might be associated with an SOI modulated by the strong perpendicular electric field. Note that the occurrence of superconductivity in 1T-SnSe₂ induced by an EDL expands the scope for 2D superconductors and further paves the way for investigating new and undiscovered superconductors.

Methods. Materials and Devices. We used the standard mechanical exfoliation method to obtain SnSe₂ flakes on 285 nm thick SiO₂ wafers. The thickness of the flakes was confirmed by a Bruker Multimode 8 atomic force microscope (AFM). By using electron-beam lithography (a pattern generation system NPGS with a scanning electron microscope FEI F50), the preselected flake was patterned into a multiterminal Hall bar configuration (as shown in Figure 1b), followed by the deposition of Ti/Au (5/50 nm) metal contacts by standard electron beam evaporation.

Transport Measurements. Following the nanofabrication, a droplet of ionic liquid (DEME-TSFI) was placed onto the device to cover the whole side gate and the rest of the sample. The device was then put into a refrigerator under high vacuum and cooled down to ~ 220 K. We expedited this process to avoid the absorption of water or oxygen from the air and the consequent electrochemical reaction. All the electronic transport measurements were carried out in an Oxford Instruments Teslatron CF cryostat. A lock-in amplifier (Stanford Research 830) was used to measure the 4-probe resistance.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.nanolett.7b05157.

Characterization of SnSe₂ samples; gate voltage V_G dependence of $R_{\text{sheet}}-T$ and analysis of the effect of ionic liquid on sample with a high gate voltage; T -dependent magnetoresistance and B -dependent $V-I$ characteristics of SnSe₂; analysis of T dependence of $V-I$ and $R_{\text{sheet}}-T$ behaviors; temperature dependence of B_{c2} in an out-of-plane and in-plane magnetic field; the modulation of magnetotransport under a perpendicular electric field (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

We would like to thank Professor Zhenhua Qiao, Yafei Ren, Dr. Peizhe Tang, and Professor Xiaoxiang Xi for fruitful discussions. This work was supported in part by the National Key Basic Research Program of China (2015CB921600, 2013CBA01603), the National Natural Science Foundation of China (61625402, 11374142, 61574076, and 11474147), Fundamental Research Funds for the Central Universities and the Collaborative Innovation Center of Advanced Microstructures. This work was partially supported by the Department of Energy, Office of Basic Energy Sciences, Division of Materials Sciences and Engineering, under contract DE-AC02-76SF00515. F.M. thanks Dimojiu Sandubashao group for experimental assistance and stimulating discussions.

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